PRACTICAL 9

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate the percentage of page hits and page faults if the system uses the Optimal page replacement technique for the page reference string entered by the user.

Example page reference string:
1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Number of frames = 3

UW PICO 5.09

```
                found = 1;
                break;
            }
        }

        // If page fault
        if(!found) {
            int farthest = i + 1;
            pos = -1;

            for(j = 0; j < f; j++) {
                int k;
                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            pos = j;
                        }
                        break;
                    }
                }

                // If page not found in future
                if(k == n) {
                    pos = j;
                    break;
                }
            }

            if(pos == -1)
                pos = 0;

            frames[pos] = pages[i];
            faults++;
        }

        // Print frames
        printf("\nFrames: ");
        for(j = 0; j < f; j++) {
            if(frames[j] != -1)
                printf("%d ", frames[j]);
            else
                printf("- ");
        }
    }

    printf("\n\nTotal Page Faults = %d", faults);
    printf("\nTotal Page Hits = %d", hits);

    float hit_ratio = (float)hits / n * 100;
    float fault_ratio = (float)faults / n * 100;

    printf("\nHit Percentage = %.2f%%", hit_ratio);
    printf("\nFault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}
```

```
(base) sakshirakhade@Sakshis-MacBook-Air ~ % nano optimal.c
(base) sakshirakhade@Sakshis-MacBook-Air ~ % gcc optimal.c -o optimal
(base) sakshirakhade@Sakshis-MacBook-Air ~ % ./optimal
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Frames: 1 - -
Frames: 1 2 -
Frames: 1 2 3
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 5
Frames: 1 2 5
Frames: 1 2 5
Frames: 3 2 5
Frames: 4 2 5
Frames: 4 2 5

Total Page Faults = 7
Total Page Hits = 5
Hit Percentage = 41.67%
Fault Percentage = 58.33%
(base) sakshirakhade@Sakshis-MacBook-Air ~ %
```